

Understanding Intent-to-Treat and Treatment-on-the-Treated Estimates

Imagine the following, admittedly silly, study.

A researcher wants to look at how much money people receive if you give them money. The answer is pretty obvious – they receive whatever you give them. But this researcher isn't very smart.

He gives 100 randomly chosen people (the experimental group) a present of \$100. He gives another 100 randomly chosen people (the control group) nothing.

The following day he surveys the sample and asks how much money they received. All the workers in the experimental group report that they received \$100 and all those in the control group report \$0. The average difference between the two groups is \$100. This is the intent-to-treat (ITT) estimate. Everyone whom the researcher wanted to give \$100, received \$100 and no one in the control group did. Because everyone who was intended to be treated actually received the \$100 and no one in the control group was treated, the ITT estimate is also the treatment-on-the-treated (TOT). Everyone who was given \$100 received \$100.

Elated by this discovery, the researcher tries a new study. He tells the experimental group that if they come to his office, they will receive \$100. He tells the control group nothing. Some of the experimental group do not bother showing up to collect the \$100, perhaps because they do not believe the researcher or think it is too much work. Let's suppose that 80% show up. The next day the researcher again conducts a survey in which he asks whether people went to his office and how much money they received. This time he finds that, on average, the experimental group received \$80 ($80\% \times \100) compared to the control group's \$0. The ITT estimate is \$80 – the people to whom he intended to give \$100 received an average of \$80 more than did the people he did not intend to treat. But, of course, the people who actually showed up got \$100 each. This is the TOT and can be calculated as $\$80/.80$ – the ITT divided by the difference in the proportion treated.

The researcher wonders whether time of day matters. In his third experiment, he tells the experimental group that they have to come to his office at midnight. As a result only 50% of the control group comes to collect the money. But word has got out that the experimenter gives out \$100 to people in the experiment who come to his office at the right time for the money. Not realizing what is happening, the researcher also gives \$100 to the 10% of the control group that show up at his office. The next day the survey reveals that the experimental group received an average of \$50 and the control group received an average of \$10. The ITT is \$40. But again, everyone who showed up received \$100. This is the TOT can be calculated as $\$40/.40$ – the ITT divided by the difference in the proportion treated.

Finally, the researcher decides to vary the amount people receive. He tells 25 experimental group members that they will receive \$25, 25 that they will receive \$50, 25 that they will receive \$75 and 25 that they will receive \$100. Again the controls are not told to collect money. And this time he is careful not to give any money to the controls. This time 30% of the experimental group show up to collect the money. When they are surveyed the next day, the average member of the experimental group turns out to have received \$21, and, of course, all members of the control group received \$0. The ITT is \$21. We no longer have an easy way to figure out the TOT. If everyone in the experimental group had collected their money, the TOT would have been \$62.50 (the average of \$25, \$50, \$75 and \$100), but we would not be surprised if people who knew they were going to be given more money were more likely to collect the money. However, we can figure out the TOT by using the formula: $TOT = ITT / (\text{difference in percentage treated})$. In this case we have $\$21/.3 = \70 . The average person who picked up the money received \$70.